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Battery pack power transfer adaptor

Abstract

A battery pack power transfer adaptor and a battery pack system that includes a battery pack adaptor. The adaptor enables a battery pack to provide high level power transfers to a variety of devices and to receive high level power transfers from a variety of power sources. The adaptor includes a battery pack interface to enable the adaptor to mechanically and electrically connect to the battery pack. The adaptor is able to transfer power at a variety of levels dependent upon the device to which it is attached.

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Background/Summary

RELATED APPLICATIONS (1) This application is a divisional application of and claims priority to U.S. patent application Ser. No. 17/237,929, filed on Apr. 22, 2021, titled, “Battery Pack Power Transfer Adaptor,” which in turn claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application No. 63/013,909, filed Apr. 22, 2020, titled “Battery Pack Power Transfer Adaptor.”

TECHNICAL FIELD

(1) This application relates to a battery pack system and a method for transferring power to and/or from the battery pack. In one implementation, the system is configured to include an adaptor for transferring power to and/or from a device coupled to the battery pack through the adaptor.

BACKGROUND

(2) Removable, rechargeable, secondary battery packs are becoming ubiquitous as more and more devices become wireless to take advantage of the advances in battery technology.

(3) Such battery packs are commonly part of cordless power tool systems and are designed and configured to operate with a variety of cordless power tool, such as drills, circular saws and grinders, for example. The battery packs and the power tools include an interface system that

enables the battery pack to couple to the power tool, as is well known in the art. Typically, these battery packs are charged using battery pack chargers that are designed and configured to charge specific battery packs. These chargers are designed and configured to plug into a wall outlet for access to alternating current (AC) mains line (utility) power or some other source of AC power, such as a generator. The ability to use the aforementioned battery packs to power unrelated devices, such as mobile phones or low power lights is also desirable.

(4) There are adaptors that may be coupled to the rechargeable battery packs that enable the battery pack to provide power to such unrelated devices.

SUMMARY

(5) The power transfer adaptor is both a tool and a charger designed to take advantage of USB 3.1's new PD (Power Delivery) specification. The feature includes the ability to supply bidirectional power up to 100 W. Because the adaptor is bidirectional, it will have many of the same properties of both a tool and a charger. For example, as a charger the adaptor can accept power up to 100 W from Type C source devices, incorporating legacy charger interfaces, standards & best practices.

And for example, as a tool the adaptor can discharge a rechargeable battery pack to charge and power various devices through its Type C Port up to 100 W and through its Type A Port up to 12 W.

(6) The adaptor leverages the Power Delivery (PD) or USB PD standard. The PD standard that allows devices to send or receive power through its Type C port (bidirectional power flow) up to 100 W and at varying voltage and current profiles. These products allow for discrete voltages of 5, 9, 12, 15, and 20V and currents up to 5 A. The PD standard uses a PD contract to establish a relationship between the PD device (the adaptor in this case) and a connected device. The contract is an agreement between two PD devices connected via Type C Cable. There are many nuances but the most basic constraints for each device are: (1) the source must establish and maintain the agreed upon voltage within $\pm 4\%$ while sourcing all the way up to the agreed upon maximum current and (2) the sink may pull any amount of current from 0 A all the way up to the maximum agreed upon current. The current may not exceed the maximum for more than a few milliseconds.

(7) A PD charger/provider/source is the device in the PD contract that agrees to send power to its partner device. A PD consumer/load/sink is the device in the PD Contract that agrees to accept power from its partner device. A PD dual role port (DRP) device is a device that is capable of being either a source or a sink depending on what is connecting to and its internal state. The adaptor is a DRP device.

(8) The adaptor can provide cordless USB "C" power delivery limited only by the available battery packs, can choose a longer run time or shorter run time battery pack to customize a user experience, and can choose a larger or a smaller battery pack size based on ergonomic preferences.

(9) These and other advantages and features will be apparent from the description and the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is an example embodiment of a power transfer adaptor.

(2) FIG. 2 is an exploded view of the power transfer adaptor of FIG. 1.

(3) FIG. 3 is an isometric view of an interior of a top housing of the power transfer adaptor of FIG. 1.

(4) FIG. 4 is an example embodiment of a battery pack system including the power transfer adaptor of FIG. 1.

(5) FIG. 5 is another example embodiment of a battery pack system including the power transfer adaptor of FIG. 1.

(6) FIG. 6 is another example embodiment of a battery pack system including the power transfer

adaptor of FIG. 1.

(7) FIG. 7 is another example embodiment of a battery pack system including the power transfer adaptor of FIG. 1.

(8) FIG. 8 is another example embodiment of a battery pack system including the power transfer adaptor of FIG. 1.

(9) FIG. 9 is another example embodiment of a battery pack system including the power transfer adaptor of FIG. 1.

(10) FIG. 10 is another example embodiment of a battery pack system including the power transfer adaptor of FIG. 1.

(11) FIG. 11 is another example embodiment of a battery pack system including the power transfer adaptor of FIG. 1.

(12) FIGS. 12a and 12b are additional example embodiments of a battery pack system including the power transfer adaptor of FIG. 1.

(13) FIGS. 13a and 13b are additional example embodiments of a battery pack system including the power transfer adaptor of FIG. 1.

(14) FIG. 14 is a simplified schematic circuit diagram of an example battery pack coupled to an example power transfer adaptor.

(15) FIG. 15 is a simplified schematic circuit diagram of an example battery pack coupled to an example power transfer adaptor.

(16) FIG. 16 is a front elevation view of a battery pack of the present invention.

(17) FIG. 17 is a right, side elevation view of the battery pack of FIG. 16.

(18) FIG. 18 is a rear elevation view of the battery pack of FIG. 16.

(19) FIG. 19 is a left, side elevation view of the battery pack of FIG. 16.

(20) FIG. 20 is a top plan view of the battery pack of FIG. 16.

(21) FIG. 21 is a bottom plan view of the battery pack of FIG. 16.

(22) FIG. 22 is a front, top, right-side isometric view of the battery pack of FIG. 16.

(23) FIG. 23 is a rear, top, right-side isometric view of the battery pack of FIG. 16.

(24) FIG. 23 is a rear, top, left-side isometric view of the battery pack of FIG. 16.

(25) FIG. 24 is a rear, top, right-side isometric view of the battery pack of FIG. 16.

(26) FIG. 25 is a front, top, left-side isometric view of the battery pack of FIG. 16.

(27) FIG. 26 is a front, bottom, right-side isometric view of the battery pack of FIG. 16.

(28) FIG. 27 is a rear, bottom, right-side isometric view of the battery pack of FIG. 16.

(29) FIG. 28 is a rear, bottom, left-side isometric view of the battery pack of FIG. 16.

(30) FIG. 29 is a front, bottom, left-side isometric view of the battery pack of FIG. 16.

DETAILED DESCRIPTION

(31) FIGS. 1-3 illustrate an example embodiment of a battery pack power transfer adaptor 10. The adaptor 10 may include a housing 12. The housing 12 may include an upper housing 14 and a lower housing 16. The lower housing 16 may include a first side housing 16a and a second side housing 16b. The upper housing 14 and the lower housings 16a, 16b may be coupled together by a plurality of fasteners 18, such as screws.

(32) The upper housing 14 and the lower housing 16 form a cavity when coupled together. The adaptor 10 may include a variety of electrical and electronic components in the cavity. For example, the adaptor 10 may include a printed circuit board (PCB) 20 including a plurality of components 22, such as resistors, capacitors, connectors, integrated circuits, etc., mounted to the PCB 20. The adaptor 10 may also include a terminal block 24 in the cavity. The terminal block 24 may include a housing 26 and a plurality of adaptor terminals 28 fixedly positioned in the housing 26. The adaptor 10 may also include a plurality of wires 30 coupling one or more of the plurality of terminals 26 to the PCB 20. The adaptor 10 may also include a belt clip 31 coupled to the upper housing 14. The upper housing 14 may also include a plurality of connector openings 32a, 32b for providing access to connectors (ports) 34a, 34b coupled to the PCB 20. The adaptor may include a

USB Type C port **34a** and a USB Type A port **34b**. In an alternate embodiment, the adaptor **10** may include a second USB Type C port instead of the USB Type A port. In another alternate embodiment, the adaptor **10** may include ports—either USB Type A, USB Type B or USB Type C—in addition to the two ports illustrated.

(33) Generally, the adaptor **10** operates with a removable battery pack. This enables the adaptor **10** to provide power to a variety of products and devices from a variety of battery packs. The adaptor **10** is able to provide bi-directional power transfer with the detachable battery pack. The adaptor may include terminals **28** to voltage tap the plurality of battery cells in the detachable battery pack and provide access to the voltage tap values. The adaptor **10** includes an automatic sleep mode in response to an undervoltage condition, an idle condition, a no cable attached condition and for various fault conditions. The adaptor **10** may also include a wake-up feature for a cable insertion condition. The adaptor **10** includes a peak current control monitor circuit to ensure that the current into or out of the adaptor **10** does not exceed pre-established values.

(34) FIG. 4 illustrates an example power tool and battery pack system. The system includes an example battery pack **100** and an example power tool **110**. The removable battery pack **100** provides power to the power tool **110**. As is well known in the art, the battery pack **100** and the power tool **110** include an interface system for coupling the battery pack **100** to the power tool **110**. This example system uses a “slide” type interface system in which the battery pack **100** includes a set of rails and grooves and a latch to interface with a corresponding set of rails and grooves and a latch recess included on the power tool **110**. Other types of interface systems, for example a “tower” type interface system could also be used. Once the battery pack **100** is coupled to the power tool **110**, the battery pack **100** can supply power to the power tool **110** for operation of the power tool **110**. Once the battery pack **100** is drained from operating the power tool **110**, the battery pack **100** can be removed from the power tool **110** and coupled to the adaptor, as described below, to charge the battery pack **110**.

(35) The adaptor **10** includes interface features similar to the power tool **110** to enable the adaptor **10** to couple to the battery pack **100**. For example, the adaptor **10** includes a set of rails **36** and a set of grooves **38** and a latch recess **40**. The system may also include a bi-directional USB Type C cord **42** include a USB Type C male connector **44a**, **44b** at each end of the cord **42**. The system may also include a plug-in power supply or rectifier **46** including a USB Type C port **48**. In this system, one male connector **44a** of the cord **42** plugs into the USB Type C port **34a** of the adaptor **10** and one male connector **44b** of the cord **42** plugs into the USB Type C port **48** of the power supply **46**. The power supply **46** plugs into an outlet coupled to an AC power source, for example, a utility mains line or a portable generator. In this manner, the adaptor **10** is able to provide a charge to the battery pack **100** from the outlet.

(36) Alternatively, the adaptor **10** may be coupled to a DC power source. Specifically, the male connector **44b** of the USB Type C cord **42** may be coupled to a USB Type C port **50** in an automobile **52** to source power from a battery or alternator in the automobile **52**.

(37) FIG. 5 illustrates another example power tool and battery pack system. The system includes a battery pack **100**, an adaptor **10**, a bi-directional USB Type C cord **42**, and a motorized electronic device **120**. The battery pack **100**, the adaptor **10**, and the cord **42** are as described above. The motorized electronic device **120** includes a motor (not shown) and an application tool, for example a grinder. The device **120** includes a USB Type C port **122**. The adaptor **10** couples to the battery pack **100**. The cord **42** plugs into the USB Type C port of the adaptor **10** and into the USB Type C port of the device **120**. The motorized electronic device **120** may (a) be without a battery and DC power from power transfer adaptor **10** drives the motor or (b) have a small battery inside the device **120** that runs the device **120** and the power transfer adaptor **10** charges the battery (either while in use or while idle—so device **120** could be operated either when connected to the adaptor **10**, i.e., corded, or when not connected to the adaptor **10**, i.e., cordlessly) or (c) have a battery which runs the DC motor and the power transfer adaptor **10** only charges the battery when the device **120** is

not in use.

(38) FIG. 6 illustrates an example computing device and battery pack system. The system includes a battery pack **100**, an adaptor **10**, a bi-directional USB Type C cord **42** and a computing device **130**, for example a tablet computing device or a mobile phone. The battery pack **100**, the adaptor **10**, and the cord **42** are as described above. The computing device **130** includes a processor and other electrical and electronic components (not shown). The device **130** includes a USB Type C port **132**. The adaptor **10** couples to the battery pack **100**. The cord **42** plugs into the USB Type C port of the adaptor **10** and into the USB Type C port of the device **130**. The computing device **130** may (a) be without a battery and DC power from power transfer adaptor **10** drives the processor and other electrical and electronic components or (b) have a small battery inside the device **130** that runs the device **130** and the power transfer adaptor **10** charges the battery (either while in use or while idle—so device **130** could be operated either when connected to the adaptor **10**, i.e., corded, or when not connected to the adaptor **10**, i.e., cordlessly) or (c) have a battery which runs the processor and other electrical and electronic components and the power transfer adaptor **10** only charges the battery when the device **130** is not in use.

(39) FIG. 7 illustrates an example non-motorized electronic device and battery pack system. The system includes a battery pack **100**, an adaptor **10**, a bi-directional USB Type C cord **42** and a non-motorized electronic device **140**, for example a head lamp or a radio. The battery pack **100**, the adaptor **10**, and the cord **42** are as described above. The non-motorized electronic device **140** includes a load such as a light or an amplifier and other electrical and electronic components (not shown). The device **140** includes a USB Type C port **142**. The adaptor **10** couples to the battery pack **100**. The cord **42** plugs into the USB Type C port of the adaptor **10** and into the USB Type C port of the device **140**. The non-motorized electronic device **140** may (a) be without a battery and DC power from power transfer adaptor **10** drives the light and other electrical and electronic components or (b) have a small battery inside the device **140** that runs the device **140** and the power transfer adaptor **10** charges the battery (either while in use or while idle—so device **140** could be operated either when connected to the adaptor **10**, i.e., corded, or when not connected to the adaptor **10**, i.e., cordlessly) or (c) have a battery which runs the light and other electrical and electronic components and the power transfer adaptor **10** only charges the battery when the device **140** is not in use.

(40) FIG. 8 illustrates another example battery pack system. The system includes a battery pack **100**, an adaptor **10**, and a motorized or a non-motorized electronic device **150**, for example a fan or a wireless speaker. The battery pack **100** and the adaptor **10** are as described above. The motorized or non-motorized electronic device **150** includes a load such as a motor—in the case of a motorized device—or a speaker—in the case of a non-motorized device and other electrical and electronic components (not shown). The device **150** includes a USB Type C connector **152**. The adaptor **10** couples to the battery pack **100**. The device **152** plugs into the USB Type C port of the adaptor **10**. The device **150** may be directly mountable onto the USB Type C port **32a** of the adaptor **10** without a cord **42**. The device **150** may include a set of rails and grooves or the weight of the device **150** secures the device **150** to the adaptor **10**. The motorized or non-motorized electronic device **150** may (a) be without a battery and DC power from power transfer adaptor **10** drives the light and other electrical and electronic components or (b) have a small battery inside the device **150** that runs the device **150** and the power transfer adaptor **10** charges the battery (either while in use or while idle—so device **150** could be operated either when connected to the adaptor **10**, i.e., corded, or when not connected to the adaptor **10**, i.e., cordlessly) or (c) have a battery which runs the motor or speaker and other electrical and electronic components and the power transfer adaptor **10** only charges the battery when the device **150** is not in use.

(41) FIG. 9 illustrates another example battery pack system. This system includes a plurality of battery packs **100** and a plurality of adaptors **10** and a plurality of USB Type C cords **42**. In this system, the adaptors **10** couple to the battery packs as described above. In a first embodiment, the

adaptors **10** include a single USB Type C port **32a**. As such, the first connector **44a** of the cord **42** is coupled to the USB Type C port **32a** of the first battery pack **100** and the second connector **44** of the cord **42** is coupled to the USB Type C port **32** of the second battery pack **100**. In this configuration, the charge of the two battery packs **100** will be balanced between the two battery packs. In an alternate embodiment, the adaptor **10** includes two USB Type C ports. In such a configuration, more than two battery packs **100** can be daisy-chained together and a first battery pack in the line of battery packs **100** can be plugged into an AC power source—as illustrated in the system of FIG. 4—to provide power to all of the battery packs **100**.

(42) FIG. 10 illustrates another example battery pack system. This system includes a battery pack **100** of a first company A and an adaptor **10** designed and configured to couple to the Company A battery pack **100** and a battery pack **100** from a second company B and an adaptor **10** designed and configured to couple to the Company B battery pack **100**. The system also includes a desktop charger **160**. The charger **160** includes a pair of USB Type C ports. The charger is designed and configured to include a plug for coupling to a supply of AC power, as described above, and to provide charging power at both USB Type C ports. The system also includes a pair of bi-directional USB Type C cords **42**. This system allows a variety of company battery packs **110A**, **110B** to be charged from the charger of one of the Company A or B or from another Company C. In this system, the first male connector **44a** of the first cord **42** couples to the USB Type C port **32a** of the Company A battery pack **100A** and the second male connector **44b** of the first cord **42** couples to a first USB Type C port of the desktop charger **160** and the first male connector **44a** of the second cord **42** couples to the USB Type C Port **32a** of the Company B battery pack **100B** and the second male connector **44b** of the second cord **42** couples to a second USB Type C port of the desktop charger **160**. As such, the desktop charger **160** can provide a charging power to both the battery packs **100A**, **100B**, either serially or simultaneously.

(43) FIG. 11 illustrates another example battery pack system. This system includes a battery pack **100**, an adaptor **10**, a cord **42** and a computing device **170** that includes a USB Type C port. The battery pack **100**, the adaptor **10**, the cord **42** and the computing device **170** couple as described above. In this example system, data regarding the battery pack may be transferred through the adaptor **10** via the USB Type C connection to the computing device **170**. The computing device **170** may adjust the charging scheme from the battery pack **100** to the computing device **170**. The computing device **170** may make these adjustments using an internal, self-operating software application or a user may make these adjustments using a user-driven application loaded on the computing device. The computing device **170** may transfer data or program code to the battery pack **100**. Alternatively, the computing device **170** may include a diagnostic application. As such, when the computing device **170** is coupled to the adaptor **10** and the adaptor **10** is coupled to a battery pack **100** that has been returned to a manufacturer, the computing device can read diagnostic or data logging information regarding the health of the battery pack **100** and any information regarding any failures in the battery pack. A user can then use the computing device **170** to determine the cause underlying the failure of the battery pack **100**.

(44) FIG. 12A illustrates another example battery pack system. This system includes a battery pack **100**, an adaptor **10**, two cords **42** and two power supplies **46**. In this system the adaptor **10** includes two USB Type C ports. The first male connector **44a** of the first cord **42** is plugged into the first USB Type C port of the adaptor **10** and the second male connector **44b** of the first cord **42** is plugged into the USB Type C port of the first power supply **46**. The first male connector **44a** of the second cord **42** is plugged into the second USB Type C port of the adaptor **10** and the second male connector **44b** of the second cord **42** is plugged into the USB Type C port of the second power supply **46**. Both power supplies **46** are plugged into a source of AC power. In this system, the battery pack **100** is supplied two times as much power than if only a single power supply were coupled to the adaptor **10** and as such, the battery pack **100** can be charged twice as fast.

(45) FIG. 12B illustrates another example battery pack system. This system includes a battery pack

100, an adaptor **10**, a single cord **42'** and a single power supply **46**. In this system the adaptor **10** includes two USB Type C ports. The cord **42'** includes three male connectors **44a**, **44b**, **44c**. The first male connector **44a** of the cord **42'** is plugged into the first USB Type C port of the adaptor **10**, the second male connector **44b** of the cord **42'** is plugged into the second USB Type C port of the adaptor **10** and the third male connector **44c** of the cord **42'** is plugged into the USB Type C port of the power supply **46**. As the power supply **46** is able to provide power to two USB Type C ports on the adaptor **10** the battery pack **100** is supplied two times as much power than if only a single USB Type C port of the adaptor **10** were being used and as such, the battery pack **100** can be charged twice as fast.

(46) FIG. 13A illustrates another example battery pack system. This system includes a battery pack **100**, an adaptor **10**, two cords **42** and two computing devices **180**, each with USB Type C ports. In this system the adaptor **10** includes two USB Type C ports. The first male connector **44a** of the first cord **42** is plugged into the first USB Type C port of the adaptor **10** and the second male connector **44b** of the first cord **42** is plugged into the USB Type C port of the first computing device **180**. The first male connector **44a** of the second cord **42** is plugged into the second USB Type C port of the adaptor **10** and the second male connector **44b** of the second cord **42** is plugged into the USB Type C port of the second computing device **180**. In this system, the battery pack **100** is able to supply charging power to two computing devices **180** and charge them both simultaneously.

(47) FIG. 13B illustrates another example battery pack system. This system includes a battery pack **100**, an adaptor **10**, a single cord **42'** and a computing device **180** with a USB Type C port. In this system the adaptor **10** includes two USB Type C ports. The cord **42'** includes three male connectors **44a**, **44b**, **44c**. The first male connector **44a** of the cord **42'** is plugged into the first USB Type C port of the adaptor **10**, the second male connector **44b** of the cord **42'** is plugged into the second USB Type C port of the adaptor **10** and the third male connector **44c** of the cord **42'** is plugged into the USB Type C port of the computing device **180**. As the battery pack **100** is able to provide twice as much charging power to the computing device **180** through two USB Type C ports of the adaptor **10** the computing device **180** is supplied two times as much power than if only a single USB Type C port of the adaptor **10** were being used and as such, the computing device **180** can be charged twice as fast.

(48) FIG. 14 illustrates a simplified schematic circuit diagram of an example battery pack **500** coupled to an example power transfer adaptor **600**. The example battery pack **500** includes a battery module (sometimes also referred to as a battery circuit) **510**. The battery module **510** includes, among other components not illustrated for purposes of simplicity but understood to those of ordinary skill in the art, a plurality of battery cells **520**. The battery cells **520** are coupled together in a first string of battery cells **530A** (sometimes also referred to as a set of battery cells) and a second string of battery cells **530B**. Each string of battery cells **530A**, **530B** includes five battery cells connected in series. In alternate embodiments, the strings of battery cells may have fewer or more battery cells. Each string of battery cells **530A**, **530B** includes a positive node A+, B+, respectively and a negative node A-, B-, respectively. The battery pack **500** also includes a plurality of battery pack terminals BT1-BT8 (sometimes also referred to as a set of battery pack terminals). The set of battery pack terminals includes a first subset of battery pack terminals BT1 and BT2 that serve as power terminals for transferring power to charge the battery cells or power to operate a device coupled to the battery pack, either directly or through an adaptor **600**. The set of battery pack terminals may also include a second subset of battery pack terminals BT3-BT8. The second subset of battery pack terminals BT3-BT8 serve as signal or data terminals for transferring low current, signals to indicate various pieces of data regarding the battery pack. The battery pack also includes an overvoltage protection (OVP) module (sometimes referred to as a circuit or controller). The OVP module includes a connection to a node between adjacent battery cells. In this manner, the OVP module can determine if a charge on each of the battery cells exceeds an overvoltage threshold. The OVP module is also coupled to the BT4 signal terminal. If one of the

plurality of battery cells exceeds the overvoltage threshold, the OVP module outputs a signal to the BT4 signal terminal. The battery pack **500** also includes an identification circuit (ID). The ID circuit is connected to the BT4 signal terminal. The ID circuit includes various components, for example resistors and/or capacitors, coupled between a voltage and a ground reference. The values of the resistors and/or capacitors determine the voltage value that is placed on the BT4 signal terminal and are indicative of various characteristics of the battery pack, for example, its chemistry, capacity, cell type, etc. The battery pack **500** also includes a thermistor circuit (TH). The TH circuit is connected to the BT3 signal terminal. The TH circuit includes various components, for example a thermistor, resistors and/or capacitors coupled between a voltage and a ground reference. The thermistor monitors the temperature of the battery cells. If the temperature of one or more of the battery cells exceed a temperature threshold, a signal is placed on the BT3 terminal. The values of the capacitor(s) in the TH circuit may also indicate various characteristics of the battery pack, for example, the number of strings of battery cells in the battery pack. In this example battery pack, the strings of battery cells **530A**, **530B** are connected together in parallel. As such, the A+ node and the B+ node are both connected to the BT2 power terminal and the A- node and the B- node are both connected to the BT1 power terminal.

(49) The adaptor **600** includes a plurality of terminals TT1-TT8 (sometimes referred to as a set of terminals). The set of adaptor terminals includes a first subset of adaptor terminals TT1 (negative power terminal) and TT2 (positive power terminal) that serve as power terminals for transferring power to charge or discharge the battery cells of a battery pack coupled to the adaptor **600**. The set of adaptor terminals may also include a second subset of adaptor terminals TT3-TT8. The second subset of adaptor terminals TT3-TT8 serve as signal or data terminals for transferring low current, signals to indicate various pieces of data regarding an attached battery pack **500**. The adaptor **600** may include a first USB type C port. The adaptor **600** may include a first USB type A port. In alternate embodiments, the adaptor **600** may include a second USB type C port instead of the USB type A port. The USB type C port may include a first positive node, a second positive node, a negative node and data node. These nodes may be formed as terminals or pins on the port. The adaptor **600** may include an ideal diode coupled to the first positive node of the type C port. The adaptor **600** may include a controllable switch coupled to the second positive node of the type C port. The first positive node and the second positive node of the type C port are coupled to the positive power terminal TT2 and the negative node of the type C port is coupled to the negative power terminal TT1.

(50) The adaptor **600** may include a first control module (sometimes also referred to as a control circuit or a controller or a microcontroller or control circuitry) and a second control module (sometimes also referred to as a control circuit or a controller or a microcontroller or control circuitry). The adaptor **600** may include a bi-directional non-inverting buck/boost converter coupled between the positive power terminal TT2 and the first and the second positive nodes of the type C port. The buck/boost converter may be coupled to the first control module. The adaptor may include a battery disconnect switch coupled between the buck/boost converter and the positive power terminal TT2. The USB type A port may include a positive node, a negative node and data node. These nodes may be formed as terminals or pins on the port. The adaptor **600** may include an ideal diode coupled to the first positive node of the type A port. The adaptor **600** may include a buck converter coupled between the positive power terminal TT2 and the positive node of the type A port. The buck converter may be coupled to the second control module. The adaptor may include an OVP module coupled to the adaptor power terminals TT1, TT2 and to the adaptor signal terminals TT5-TT8. The OVP module may be coupled to the second control module. The adaptor **600** may include a first thermistor circuit and a second thermistor circuit to monitor the temperature of the adaptor **600** and its components. The first and second thermistor circuits may be coupled to the second control module.

(51) The first control module may be coupled to the battery disconnect switch to control the battery

disconnect switch based on various signals received by the first control module. The first control module may be connected to the TT4 signal terminal. The second control module may be connected to the TT3 signal terminal. The second control module may be coupled to the control switch coupled to the second positive node of the type C port. The second control module may control the control switch coupled to the second positive node of the type C port based upon one or more signals received by the second control module.

(52) After a user inserts a source into the adaptor Type C port, the adaptor performs a negotiation with the partner device. When negotiation is complete, a PD contract is established wherein the partner device Type C source is obligated to provide the contract voltage and our unit is obligated to not exceed the agreed upon maximum current level. Typical voltage and current levels are as follows:

(53) TABLE-US-00001 Voltage Level Current 5 V 1 A, 1.5 A, 3 A 9 V 1 A, 1.5 A, 3 A 12 V 1 A, 1.5 A, 3 A 15 V 1 A, 1.5 A, 3 A 20 V 1 A, 1.5 A, 3 A, 5 A

(54) The adaptor can read the PD contract information and adapt its current intake accordingly.

(55) Current Level Selection

(56) When selecting a current level for charging, the adaptor will consider and comply with: the partner device (charger's) maximum current output level (Ibus limit) and the battery's maximum safe allowable current level (Ibat limit). The adaptor's software is perpetually monitoring both the PD/Type C activity to determine the Ibus limit and battery/ambient conditions to determine the Ibat limit.

(57) Partner Device & Ibus Limit

(58) As mentioned above, the adaptor will attempt to pull current commensurate with the maximum current the partner device can supply minus about 5%. This ensures we minimize charge time but do not exceed the PD contract conditions.

(59) Pack ID Limits

(60) The adaptor reads and considers both the ID Resistor (ID line to ground) and the ID Capacitor (NTC line to ground) of the battery pack with independent microcontrollers and chooses the lower of the two values to select the maximum amount of current the battery pack can take when charging the battery pack.

(61) Charger NTC & Thermal Foldback

(62) In order to prevent the adaptor from overheating, there is a thermal foldback routine. To do this we read the two on-board NTCs and adjust the battery current accordingly. The routine is as follows: at 115° C. we reduce the maximum allowable battery current from 4 A or higher to 2 A maximum, at 95° C. we re-allow the maximum current to be 4 A or higher (pack dependent), at 125° C. we disallow current flow entirely, at 120° C. we re-allow current flow of 2 A.

(63) Pack NTC Limitations

(64) In order to protect the attached battery and comply with charging schemes for the attached battery pack, the adaptor must react to the pack NTC conditions as follows: below 0° C., allow 0 A charging, between 0° C. and 10° C., allow 2 A maximum charging, above 60° C. allow 0 A charging; each options including a few degrees of hysteresis.

(65) Over-Voltage Protection Reset Limits & Topping Off

(66) The adaptor uses top-off “step down” charging schemes similar to charger. The rule base is as follows: for bulk charge (0 OVP trips) pull the maximum amount of current allowed by all the other rules (CNTC, Pack, Ibus, Pack ID); after 1 OVP Reset: set the limit to 2 A; after 2 OVP Resets: set the limit to 1 A. For all subsequent resets, set the limit to 300 mA as a maintenance charge. However, if a Type A device is attached, the minimum current will be 2 A instead of 300 mA in order to ensure that adaptor charges the attached battery pack at a rate that is faster than the Type A discharge rate.

(67) Type C Sourcing/Discharging

(68) After a user inserts a sink into the adaptor Type C port, the adaptor performs a contract

negotiation with the partner device. When the contract negotiation is complete, a PD contract is established wherein the partner device Type C sink is obligated to pull less than the PD contract maximum current and the adaptor is obligated to provide the requested voltage within $\pm 5\%$. If the battery pack NTC reading indicates a reading greater than 70°C . the adaptor will not allow discharge through the Type C port. If one of the adaptor NTCs detects a temperature greater than 120°C . the adaptor will not allow discharge through the Type C port. If the adaptor detects a stack voltage of $<15\text{ V}$ it will cease to allow discharge on the Type C port. The adaptor will only re-enable discharge once the stack voltage exceeds 17.8 V .

(69) Type A Sourcing/Discharging

(70) After a user inserts a Type A device into the Type A port of the adaptor, the adaptor detects the presence of a load and turns on the Type A bus within $\sim 1\text{ s}$. If the adaptor temperature or the battery pack temperature falls below -20°C . the adaptor will stop current flow through the Type A port. If the adaptor detects a stack voltage of $<15\text{ V}$ it will stop discharge on the Type A port. The adaptor will only re-enable discharge once the stack voltage exceeds 15.5 V . The Type A bus will remain on for at least 8 hours. At that point the adaptor checks whether a device is still inserted. If a device is still inserted to the Type A port, the adaptor keeps the bus on. If a device is not still inserted to the Type A port, then the adaptor turns off the Type A port.

(71) Numerous modifications may be made to the exemplary implementations described above. These and other implementations are within the scope of this application.

Claims

1. A battery pack power transfer adaptor, comprising: a USB type C port; a set of power terminals including a positive power terminal and a negative power terminal, the set of power terminals configured to mate with a rechargeable battery pack for transferring power to and from the battery pack; the positive power terminal coupled to the USB type C port at a first node and at a second node of the type C port and the negative power terminal coupled to the USB type C port at a third node; a first control module and a second control module; and a bi-directional non-inverting buck/boost converter coupled between the positive power terminal and the first and the second node of the type C port.
2. The battery pack power transfer adaptor, as recited in claim 1, further comprising an overvoltage protection module coupled to the adaptor positive power terminal and the adaptor negative power terminal and to a plurality of adaptor signal terminals.
3. The battery pack power transfer adaptor, as recited in claim 2, wherein the overvoltage protection module is coupled to the second control module.
4. The battery pack power transfer adaptor, as recited in claim 1, further comprising a first thermistor circuit and a second thermistor circuit to monitor the temperature of the adaptor and its components.
5. The battery pack power transfer adaptor, as recited in claim 4, wherein the first and second thermistor circuits are coupled to the second control module.
6. The battery pack power transfer adaptor, as recited in claim 1, wherein the buck/boost converter is coupled to the first control module.
7. The battery pack power transfer adaptor, as recited in claim 1, further comprising a battery disconnect switch coupled between the buck/boost converter and the positive power terminal.
8. The battery pack power transfer adaptor, as recited in claim 1, further comprising a USB type A port, the USB type A port including a positive node, a negative node and data node formed as terminals on the port.
9. The battery pack power transfer adaptor, as recited in claim 8, further comprising a buck converter coupled between the positive power terminal and the positive node of the type A port.

10. The battery pack power transfer adaptor, as recited in claim 9, wherein the buck converter is coupled to the second control module.
